

Why no hyperparameter search rescues honest market making

Re: "there must be a parameter combination where ML/RL succeeds, at least in-sample"

1. We ran exactly that experiment (exp 58, Contribution 30)

Same algorithm, same data, same epoch budget, train=eval (explicitly licensed to memorize in-sample). The **only** thing that changes is whether the fill model grants free queue priority:

Arm	Setup	Result
Control (artifact)	TabularQ, 19 actions (inside spread allowed), no-queue fill model, 200 epochs, train=eval	+\$58/day (best epoch +\$88)
Honest	same algorithm & budget, 63 actions (at-touch / outside only), real L2 queue model	best stable +\$0.7/day
Honest, DQN	continuous state, same honest fill model	converges to ~0 fills/day

Architecture, epoch budget and action granularity were all already varied. What flips +\$58/day to +\$0.7/day is one binary switch: free queue priority or not.

2. "In-sample" was the whole point of the design above

Train=eval, 200 epochs, free to memorize -- still can't clear ~\$1/day honestly. In-sample doesn't help because the honest reward, summed over the dataset, really is ~0 (the "why" is in section 6).

3. The absolute ceiling: even an omniscient oracle (Contribution 34)

Drop hyperparameters entirely. Take an honest at-touch quoter's real fills and keep, in hindsight, only the ones that turned out favorable:

Horizon	Honest (no foresight)	Oracle (perfect foresight)
1 s	\$0.60/day	\$19.76/day
30 s	\$4.06/day	\$30.03/day

Ceiling ~ \$20-30/day over **every** policy/architecture, granting literal omniscience -- still below the +\$45-88/day the broken fill model hands a trivial agent for free.

4. Better ML doesn't move it either (Contribution 31)

XGBoost on 8 microstructure features (OFI, momentum, OBI, vol, ...), OOS AUC = 0.75 for direction -- genuine skill. Yet on the per-trade \$ metric it is **marginally worse** than plain order-book imbalance, and neither clears trading costs. More capacity buys classification accuracy, not P&L.

5. Not a search / code problem (Contribution 33)

Same engine + same closed-form A-S/GLFT formulas, fed a synthetic market with a known edge: find it to the dollar, zero search -- expected \$770.40 vs realized \$769.44. When profit exists, nothing here needs tuning to find it.

6. The theoretical "why"

Glosten-Milgrom / Wyart-Bouchaud zero-profit equilibrium: competitive entry prices the spread/queue so the marginal honest maker's expected profit is exactly zero (half-spread* ~ $\sigma_{\text{per trade}}$, $\kappa \sim 1/\sigma$). Not a property of this dataset -- it's the definition of a competitively-priced liquidity market. Searching for a hyperparameter combo that's profitable is like searching for a rule that beats a fairly-priced option on a liquid underlying.

Bottom line

	Broken fill model (free queue priority)	Honest fill model (real L2 queue)
Classical A-S / GLFT	profitable	~0 / negative
RL, 200 epochs, train=eval	+\$45-88/day	+\$0.7/day
Oracle (perfect foresight)	n/a	\$20-30/day

One variable -- queue priority -- explains every cell. Algorithm and hyperparameters don't move it.